NATIONAL UNIVERSITY



Third Year Syllabus
Department of Soil Science

Four Year B.Sc. Honours Course
Effective from the session : 2013–2014

National University

Subject: Soil Science

Syllabus for Four Year B.Sc. Honours Course

Effective from the session: 2013-2014

Year wise Papers and marks distribution.

THIRD YEAR

Paper Code	Paper Title	Marks	Credits
233301	Pedology	100	4
233303	Soil Physics-II	100	4
233305	Soil Chemistry	100	4
233307	Soil Microbiology	100	4
233309	Soil Fertility and Plant Nutrition	100	4
233311	Environmental Pollutions and Waste Management	100	4
233313	Soil and Plant Biochemistry	100	4
233314	Practical	100	4
	Total =	800	32

Detailed Syllabus

Paper Code	233301	Marks: 100	Credits: 4	Class hours: 60 hrs.
Paper Title:	Pedology			

Scope of Pedology. Pedological and edaphological approaches in the study of soil; Concept of soil as a natural system.

Concepts of soil formation; Soil horizons-formation of soil horizons; Master horizons and diagnostic horizons in soil; weathering and soil formation; Pedochemical weathering processes. Detailed study of the factors of soil formation; soils in relation to landscape; soil catena.

Geographical distribution of soils in the world**: Z**onality concept of soils. Major climatogenic and climatogenically subdued soils- their genesis, properties and importance; Study of the orders and suborders of Soil Taxonomy; environmental settings, genetic processes, properties and uses of soil orders. Characteristics of major soil groups of the world.

General conditions of soil formation prevailing in Bangladesh; dominant soil forming processes; characteristics of major soil groups. Soil characteristics of the three major landscapes of Bangladesh. Agro-ecological Zones (AEZ) of Bangladesh. Upazilla land & Soil resource user nirdeshika.

Books Recommended :

1.	Soil Genesis and Classification	:	Boul et.al. 4th. ed. 1997.
2.	A text book of Pedology	:	*J.* Sehgal, Kalyani Publisher. New Delhi**.** - 3rd ed. 2011
3.	Handbook of Pedology	:	Dokuchaev
4.	Upazilla land & Soil resource user nirdeshika	:	SRDI.
5.	SOILS: Their survey & Taxonomic classification	:	SRDI-2013

Paper Code	233303	Marks: 100	Credits: 4	Class hours: 60 hrs.
Paper Title:	Soil Physics - II			

The Solid phase: Soil structurc – genesis, evaluation, management and agricultural significance. Soil consistence -Atterberg constants and their practical significance.

The Liquid phase: Water retention in soil - molecular properties of water, fluid properties of water, properties of water near particle surfaces. Soil water content - definition and measurement; Energy state of water in soil: soil water potential, measurement of soil water potential; water retention characteristics; hysteresis.

Saturated and unsaturated flow of water; Field capacity concept; Available water; Factors affecting water storage in soil profile; Field water balance; Infiltration; Factors affecting infiltration; Methods of measuring infiltration rate; Evaporations; Evapotranspiration (ET) and consumptive use (CU); Factors affecting Evaporation and Evapotranspiration.

Soil air: composition of gaseous phases: variation of air composition under varied conditions; effects of aeration on chemical and biological processes.

Soil water-plant relations; the soil-plant-atmosphere continuum (SPAC); Absorption of water by plant roots.

Books Rccommanded.

1.	Fundamentals of Soil Physics	:	Daniel Hillel.
2.	Soil Physics	:	Willam A Jury.

3. Soil Physics : Helmut Kohnke
4. Introduction to Environmental Soil Physics : Daniel Hillel
5. Methods in Soil Physics : Sk. Jalota.
6. Soil Physics : R.P. Tripathi

Paper Code	233305	Marks: 100	Credits: 4	Class hours: 60 hrs.
Paper Title:	Soil Chemistry			

Concept and scope of soil chemistry; chemical aspect of the three phases of soil

Soil inorganic matter: their nature and general properties.

Humus - its origin, functional groups, and various hypotheses of humus formation; humus fractionation, characteristics and importance.

Soil solution - its concentration and composition, factors controlling them: dynamics of soil solution, movement of salts in soil; chemical equilibrium, importance of soil solution in soil development and plant growth.

Ion exchange in soil*:* Origin of ion exchange properties; factors influencing ion exchange. Cation, anion and ligand exchange; ion exchange capacities of various soil constituents; importance of ion exchange in Agriculture.

Soil reaction: Active, reserve, and potential acidity in soil; measurement of soil acidity: sources of hydrogen and hydroxyl ions in soil, importance of pH on soil properties and plant growth.

Chemistry of waterlogged soil - Characteristics of waterlogged soils; electrochemical changes; transformations of N, P, S, Fe, Mn and other nutrients; chemical changes during drying of waterlogged soils; decomposition of organic matter and its effects on soil chemical properties; chemical, physical and biological conditions prevailing in paddy soils; adaptation of plants to waterlogged soils.

Books Recommended:

1. Principles of Soil Chemistry : Kim H. Tan.
2. Soil Chemistry : Bohn. McNeal, A. D. connor.
3. Chemistry of the Soil : F.E Bear, Oxford IBH Publishing Co Pvt. New Delhi.
4. Soil Chemistry and its application : Malcolm Cresser& Ken Kilham & Tony Edwards. Cambridge University press
5. Principles of Soil Science : M. M. Rai
6. Chemistry of Sube\merged Soil – F. N. PonamPerruma

Paper Code	233307	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	Soil Microbiology			

Soil micro flora: Bacteria, Actinomyceties, Fungi and Algae - their structure, reproduction and simple classification. Growth and reproduction of bacteria-growth phases, growth measurement. synchronous and asynchronous growth. Most probable numbers (MPN). Counting techniques.

Microorganisms and variours environmental stress on extreme environments, microbial communities and their adaptations.

Biological nitrogen fixation (BNF): microbiology and biochemistry of nitrogen fixation: nitrogen fixation under different environmental conditions, preparation and use of inoculum; adaptation and success of bio-fertilizers; impact of bio-fertilizer on agriculture; measurement of BNF.

Mycorrhiza: definition and classification of mycorrhiza: role of ecto-and endo-micorrhiza in crop production.

Azolla- Anabena symbiosis -its physiology and uses in tropical agriculture. Role of microorganisms in soil fertility and plant growth.

Interactions among single and diverse microbial populations neutralism. commensalism, synergism, mutualism, parasitism and predation

Books Recommended:

1. Introduction to soil microbiology : Martin Alexander & John, 2nd'ed.
2. Soil microbiology : Robert L Tate, 2nd'ed
3. Fundamentals of Microbiology : Martin Frobisher W.B.saunders company Phiadelphia/London/ Toronto.
4. Soil Microbiology : N. S. Subba Rao-2011
5. Soil Microbiology : R. R. Mishra-2010

Paper Code	233309	**Marks: 100**	**Credits: 4**	**Class hours: 60 hrs.**
Paper Title:	Soil Fertility and Plant Nutrition			

Past and recent advances in the field of soil fertility and plant nutrition: factors affecting plant growth and development.

Plant nutrient elements: essentiality of nutrient elements, macro and micronutrients: mechanism of nutrient uptake in plants; physiological functions of various plant nutrient elements.

Fertilizer materials: sources and manufacture of major fertilizer materials; single, compound and mixed fertilizer; fertilizer grades and fertilizer ratios; fertilizer law; organic fertilizer and bio-fertilizer.

Diagnosis of fertilizer needs: different approaches to diagnosis; visual symptoms; soil and plant analysis; biological methods and laboratory, green house and field experiments.

Methods of fertilizer application; fertilizer movement in soil; fate of added nutrient in soil; residual effects of fertilizers; salt index; balanced fertilization; effect of fertilizer application on soil quality and environment.

Forms and labile pool of plant nutrients in soil; nutrient buffering capacity of soil; quantity-intensity relationships. Nutrient fixation-mechanisms and factors controlling the non-biological fixation of N, P and K in soil; fate of fixed nutrients in soil.

Nutrient interactions**;** antagonistic-synergistic activity, fertilizer response in plants; mathematical models to express fertilizer response; various equations and curves; calculation of fertilizer needs of soil.

Nutrient status of Bangladesh soils; AEZ and fertilizer requirement; fertilizer application in a single crop and in a crop sequence; crop modeling.

Books Recommended

1. Soil Fertility and Fertilizer : Tisdale & Nelson
2. Soils & Soil Fertility : L. M. Thompson & F. R. Troch
3. Soil Fertility : Henry D Foth.
4. Soil Fertility : Millar.
5. Soil Fertility Management for Sustainable Agriculture. : R. Prasad.
6. Principles of Plant Nutrition : K. Mengel and E,A, Kirkby . Panima Publishing Co. New Delhi. India

Paper Code	233311	**Marks: 100**	**Credits: 4**	**Class hours: 60 hrs.**
Paper Title:	**Environmental Pollutions and Waste Management**			

Soil environment: meaning of soil environment; environmental awareness; concept of environmental pollution.

Sources of soil pollution: sewage sludge, industrial wastes and effluents, hospital wastes, city wastes. agrochemical, PCBs, Organometallic compounds, fossil fuels, radioactive materials and microbiological pollutants. Effects of pollution on soil ecosystem. Effluents and city wastes - their composition (metal contents) and nutrient value; permissible limit values of heavy metals.

Heavy metals: sources and world production, contents of heavy metals in normal soil, water and plants; permissible limits in soils, water and crops; biochemistry, chemical toxicology and impact of heavy metals on soil ecology, crops and humans; LD50 values; remediation of heavy metal toxicity.

Radioactivity: sources of natural and artificial radioactivity; hazards caused by radioactivity; impact of radioactivity on human and animal management and disposal of radioactive wastes.

Waste management principles: Types, collection. transportation, treatment, resources recovery- and disposal of various solid wastes. Principles and practices of bioremediation of solid wastes.

Books Recommended

1. Environment Pollution : Katyal
2. Environmental Hazards & Human Health : Richard B.Philp,Lewis Publishers.
3. Hand Book of Ecotoxicology : Peter Calow.

Paper Code 233313 **Marks: 100 Credits: 4 Class hours: 60 hrs.**

Paper Title: Soil and Plant Biochemistry

Biota and biomass of soil fractions. Soil as a catalytic system. General principles of catalysis-definition, classification, characteristics. Catalysis in soil, biotic and abiotic reactions, enzymes as catalysts in soil, enzyme activity in soil, free and immobilized enzymes, enzymes active on particulate substrate, classification of enzyme, determination of enzyme activities in soil. Factors affecting enzyme activity in soil.

The classical Michaelis - Menten equation. Decomposition of organic residues in soil-influence of residue and soil properties. Constraints on biological activity, microbial and enzymic diversity, activity and persistence of extra-cellular enzymes. Clay-enzyme and humus-enzyme complexes. Interaction between soil minerals and microorganisms. Enzymic pathways for the degradation of cellulose, hemicellulose, lignin and pectin. Mineralization of C , N and S; Nitrification and Denitrification.

Carbohydrate and protein synthesis in plants - enzymic pathways, energy relationships and involvement of trace metals. Respiratio glycolysis, Kreb's cycle & electron transport system, Nucleic acids-composition, classification and functions; Replication of DNA, transcription and translation of RNA. Hormones, vitamins and plant growth regulators.

Books Recommended :

1. Soil Microbiology **:** N.S. Subba - 2011
2. Fundamentals of Biochemistry : J. L. Jain, S. Jain & N. Jain - 2011
3. Introductory plant Physiology : G. Ray Noggle & Geroge J. Fritz. Prentice Hall of India Pvt. Ltd. New Delhi
   : Devlin

Paper Code	233314	**Marks: 100**	**Credits: 4**	**Class hours: 60 hrs.**
Paper Title:	**Practical**			

1. Study of soil site characteristics: topography, drainage, parent material and vegetation; detailed study of soil profiles in a catena; collection and preservation of soil samples for laboratory analysis.
2. Determination of active and reserve acidity of soil **;** determination of lime requirements of soil; determination of total and available N P K and S and micronutrients in soil: determination of EC. Soluble salts, CEC and exchangeable bases; estimation of BSP and ESP; determination of the effects of acidification and liming on soil properties.
3. Preparation of culture solution; sand culture experiment to observe the effects missing elements on some common crops; acquaintance with different fertilizers and methods of their use; choice of extraction methods to assess the availability of N, P, K. S, Ca, Fe and Zn; analysis of plants for different nutrients .
4. Microbiological tools and techniques, preparation of media, counting of bacteria, simple staining techniques. Identification of bacteria by biochemical tests and similarity matrix; determination of ammonification and nitrification potential of soil; study of nitrogen fixation by free living bacteria; preparation of inoculum and its application; measurement of BNF; sterilization of seed and soil.
5. Determination of water quality for drinking and irrigation purposes.
6. Estimation of microbial biomass, carbon, nitrogen and phosphorus.

Books Recommended

1. A. Handbook on Analysis of Soil, Plant & Water- S. M. Imamul Huq & Md. Didar ul Alam
2. Methods in Environmental Analysis – Soil, Water & Air- P.K. Gupta